Program 1:
Build a Docker Container from a custom Dockerfile
31/10/2025

Project Directory:

prog1
- |- app.py
- |- requirements.txt
- |- Dockerfile

Program Code and steps for execution:

Step 1: Move to the directory where we work for DevOps lab. Then create a project directory with the name prog1
> mkdir prog1
> cd prog1/

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog1
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog1
```

Step 2: Inside the prog1 folder there will be 3 files
1. requirements.txt 2. app.py 3. Dockerfile
All the above files contains the code which is in below pictures respectively

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ sudo nano requirements.txt
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ cat requirements.txt
flask
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ sudo nano app.py
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ cat app.py
from flask import Flask

app=Flask(__name__)

@app.route("/")
def hello():
        return "Hello, Docker!!!"

if __name__=="__main__":
        app.run(host="0.0.0.0", port=5000)
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ sudo nano Dockerfile
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ cat Dockerfile
### Dockerfile

# This is the Dockerfile which contains the information to build the docker container images

# Use the slim python as a base image
FROM python:3.9-slim

# Set the working directory for the container
WORKDIR /app

# Copy the text/content from the requirements.txt and install all the dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy the application code
COPY . .

# Set the flask environment variables
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

# Expose the port that flask will run an app on
EXPOSE 5000

# Define the commands to run the flask application
CMD ["flask", "run"]
```

Step 3: The final Project directory looks like the below

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ ls
app.py  Dockerfile  requirements.txt
```

Step 4: Build the docker image from the Dockerfile of prog1 using the command,
> docker build -t prog1 .

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ docker build -t prog1 .
[+] Building 2.3s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 710B
 => [internal] load metadata for docker.io/library/python:3.9-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => [internal] load build context
 => => transferring context: 269B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:fdcb9ac48df4e541b1800968ec4077e4c8fb81b468b32c90d4730e8254e01e84
 => => exporting config sha256:6de5de28faba26fb5b8c79b981e16cd4a02a9a35e370dc2d94c93b10a56c388a
 => => exporting attestation manifest sha256:52188d73cc8c060abf971e9401b5fb4565d20705aa402fd11afacdeaf835a2f0
 => => exporting manifest list sha256:d10a716af680c988678c93a054841ed329105cd5614b2c6f6b959bcf6438019b
 => => naming to docker.io/library/prog1:latest
 => => unpacking to docker.io/library/prog1:latest
```

Step 5: Run the image which has been built by using the command,
> docker run -d -p 5000:5000 –name flask-container prog1

Step 6: Check for running containers

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ docker run -d -p 5000:5000 --name flask-container prog1
c730a91d8013f5f7418d1461fba12635c3e1a0a30e7b0dcf2f6ff6302c4276fa
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS                                       NAMES
c730a91d8013   prog1     "flask run"   6 seconds ago   Up 5 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
```

Step 7: Open any browser in the local machine and search for,
> http://localhost:5000/

The output will be as show below

Step 8: Stop and remove all the running container, also remove the created image by using the command,
> docker container stop <4_characters_of_container_id>
> docker container rm <4_characters_of_container_id>
> docker rmi <4_characters_of_image_id>

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ docker container stop c730
c730
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ docker container rm c730
c730
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ docker images

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
prog1:latest   d10a716af680        200MB         48.8MB
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog1$ docker rmi prog1:latest
Untagged: prog1:latest
Deleted: sha256:d10a716af680c988678c93a054841ed329105cd5614b2c6f6b959bcf6438019b
```